Regime-Dependent Feature Importance in Oil Price Movement Using Logistic Regression

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1 Abstract

Oil price movements are often influenced by numerous factors, including financial market conditions and macroeconomic events. This study investigates the importance of different features for predicting oil price movements in different volatility periods using logistic regression. Three different logistic regression models meant to predict up and down movements of oil prices were trained with data from time periods with different volatility levels and financial crises and were tested with the same 7 engineered features and 3 raw market features. The raw market features were derived from the close prices of WTI crude oil futures price, CBOE Volatility Index (VIX), and the SPY 500 ETF Trust. The weights of each feature per model were analyzed and ranked by their importance in the model's predictions. The results show that the model had the most influence from SPY-related features when predicting the movements of oil prices in times of crisis and stability, while focusing more on features derived from the WTI crude oil futures price in periods of medium volatility. These results highlight the changing relationship between stock market movements and oil price behavior, particularly during periods of market shock.

2 Introduction

Oil prices play a significant role in the global economy and are influenced by a variety of macro-economic and financial factors. In particular, relationships between oil prices and financial markets such as equity indices and volatility measures can change significantly during macroeconomic shocks and disruptions. An example of this occurred during COVID-19, where equity indices, volatility measures, and the price of oil measured sharp and unusual co-movements.

However, predicting oil price movements has been shown to be difficult due to sudden shocks and non-linear relationships with financial markets. Financial markets often experience periods of stability and instability, during which relationships between variables can change drastically. These relationships make it difficult for traditional models to predict the movements of oil prices.

Some existing models assume linearity in the relationships between financial markets and oil prices. However, shocks and market disruptions can interfere with the relationships between variables. This motivates the need to understand how different variables relationships behave in periods of crises and stability, so that models can use more meaningful features for predictions depending on how the stability of the market is at that moment.

This study implements three separate logistic regressions in order to predict the upward and downward movements of oil prices while being trained on different time periods of crisis, stability, and medium volatility. The raw data collected included the close prices of the WTI crude oil futures price, VIX indicators, and SPY indicators. Each model was trained and evaluated with the same 10 features. These features were then ranked based on their importance to the model's predictive ability by comparing the absolute value of their weights.

3 Data

This study uses financial market data from multiple sources to analyze oil price movements under different volatility regimes. The data used were VIX indicators, SPY indicators, and the WTI crude oil futures price. The data came from Yahoo Finance, from three separate periods during three levels of volatility: stable regime data from January 2015 to December 2017, crisis regime data from February 2020 to June 2020, and medium volatility regime data from January 2018 to December 2019.

After the raw data was collected, the data from the different indicators of financial markets and oil prices were aligned in time series and cleaned to allow clean merging between the indicators and oil prices. After the indicators were cleaned up, they were merged into one dataset per regime, which would be used for further feature engineering.

The main factor used to find these time periods came from analyzing current events and shocks in the market. The period in which the crisis model was trained was characterized by the rise of the COVID-19 epidemic, which had caused recessions and the crash of global stock markets. The period in which stable regime model was trained was characterized by the oversupply of crude oil, which gradually recovered. The period in which medium volatility model was trained with was characterized by a large recovery from the earlier crashes in 2015-2016, but there were still trade disputes between the USA and China.

4 Feature Engineering

Financial markets have often been characterized by the noise in their movements, making prediction a hard task for a model. However, the task can be made simpler by engineering features that can help the model to learn long and short term trends, like moving averages and market lags. Because of this, feature engineering is commonly used in financial forecasting to help models learn patterns from more meaningful data along with the raw prices themselves.

In the logistic regression models used in the study, lag features were used as engineered features. These features are defined as follows:

$$x_{t-k} \tag{1}$$

where x_t represents the value of a financial variable at time t, and k denotes the lag length (e.g., $k = 1, 5$).

In addition to lag features, moving averages were constructed to smooth short term fluctuations in data. These are defined as follows:

$$MA_n(t) = \frac{1}{n} \sum_{i=0}^{n-1} x_{t-i} \tag{2}$$

where $MA_n(t)$ represents the moving average at time t, n is the window size, and x_{t-i} denotes the value of the variable at time $t - i$.

These features were applied to all raw financial variables, including the WTI crude oil futures price, VIX indicators, and SPY indicators, resulting to a total of 16 engineered features. Table 1 summarizes the final set of engineered features in this study.

Feature	**Description**	**Window/Lag**
oil	Current oil price	–
spy	Current SPY value	–
vix	Current VIX value	–
oil_lag_1	Oil price lag feature	1 day
oil_lag_5	Oil price lag feature	5 days
spy_lag_1	SPY lag feature	1 day
vix_lag_1	VIX lag feature	1 day
oil_ma_5	Oil price moving average	5 days
oil_ma_20	Oil price moving average	20 days
spy_ma_20	SPY moving average	20 days

Table 1: Raw and engineered features used in the logistic regression models.

5 Methodology

Logistic regression was selected as the primary model for this study due to its ability to perform binary classification and provide interpretable feature coefficients. Since the primary objective of this study was to analyze feature importance rather than maximize predictive performance, logistic regression was chosen due to its interpretable coefficient structure and its simplicity.

Before producing a probability, logistic regression calculates a score based on the weighted sum of the input features. Let z be denoted as the linear combination of input features:

$$z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n \tag{3}$$

where z is the linear predictor, β_0 is the intercept, β_i are the model coefficients, and x_i are the input features.

This value is then passed through the sigmoid function:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-z}} \tag{4}$$

where $P(y = 1 \mid x)$ represents the probability of an upward movement in oil prices, and z is the linear predictor defined in the previous equation.

Three separate logistic regression models were trained and evaluated, each corresponding to a different market regime. All models used the same set of engineered features to ensure comparability across regimes.

After training, the linear regression model coefficients were compared to analyze feature importance throughout the regimes. The magnitude of each coefficient was used as the comparison baseline. In order to truly analyze what the model was most influenced by, the absolute value of the model coefficients were used in comparison.

6 Results

This section presents the feature importance results obtained from the logistic regression models trained on different market regimes. Features were compared based on the absolute value of the model coefficients.

In the crisis regime, the absolute value of the weights showed that the current SPY price was the most influential to the model's predictions. However, the actual weight itself the model used was negative, suggesting that the current SPY price has an inverse relationship to oil prices in crisis regimes. Table 2 shows the top five most important features for predictions in crisis regimes.

Feature	Weight	Absolute Weight
spy	-0.964	0.964
oil	-0.802	0.802
oil_lag_5	0.585	0.585
vix_lag_1	-0.565	0.565
spy_ma_20	0.552	0.552

Table 2: Feature weights for the crisis regime logistic regression model.

In the stable regime, the SPY 20 day moving average had the highest absolute weight. The actual model weight was also positive, suggesting that SPY based features matter more in predictions of oil price movements in stable regimes. Table 3 shows the top five most important features for predictions in stable regimes.

Feature	Weight	Absolute Weight
spy_ma_20	0.853	0.853
spy_lag_5	-0.806	0.806
vix_ma_5	0.676	0.676
vix_lag_1	-0.553	0.553
oil_ma_20	-0.342	0.342

Table 3: Feature weights from logistic regression model in stable regimes

In the medium volatility regime, the raw price of oil and other oil based features were the most influential to the predictions made by the model. However, the actual weight the model assigned to the current oil price as negative, suggesting that oil based features provide more predictive signal during medium volatility regimes. Table 4 shows the top five most important features for predictions in medium volatility regimes.

Feature	Weight	Absolute Weight
oil	-1.126	1.126
oil_lag_1	0.809	0.809
spy_lag_1	-0.646	0.646
oil_lag_5	-0.512	0.512
oil_ma_5	0.433	0.433

Table 4: Feature weights for the medium volatility regime logistic regression model.

7 Discussion

The results show that feature importance varies significantly across different market regimes. These results suggest that the relationship between oil price and financial market variables do not have a constant relationship. Because of this, models trained under a certain regime may have difficulty predicting the movement of oil prices in another regime.

In the crisis and stable regime, SPY related features had the highest feature importance in the crisis and stable regime models. These results suggest that equity market features yield a significant amount of influence on models trained during these regimes. However, the actual weight the model assigned to these features was negative, further suggesting that the model interpreted SPY related features to have an inverse relationship to the movement of oil prices.

However, during the medium volatility regime, crude oil based features had the highest feature importance in the model. These results suggest that the model interpreted that medium volatility markets are commodity-driven, meaning that indicators such as SPY and the CBOE Volatility Index had little influence in the predictions made by the model. Out of the top five features with the highest importance, only the SPY 1 day lag feature was on the list.

8 Limitations

This study contains multiple limitations that must be taken into consideration. For example, feature weights in models do not reflect causation, but merely association. The logistic regression models used in this study also may not have captured the non-linearity in financial and oil markets, meaning that a more sophisticated model could achieve higher financial forecasting. Also, the features engineered in this model do not take into account other macroeconomic variables that may influence oil prices.

9 Conclusion

This study investigated the importance of financial market features in predicting oil price movements across different market regimes. During these regimes, the relationship between financial and oil price variables changed drastically. In crisis and stable regimes, SPY based indicators showed to have dominant feature importance, while crude oil based features dominated in medium volatility regimes. These results suggest that predictive features behind the movements of oil prices are defined by market regimes and are not linear.